DANIEL NG

604-727-3233 | daniel.whn@outlook.com | danielng.pages.dev | linkedin.com/in/danielngEE | github.com/danielwhn25

Summary

Firmware and software test engineer with hands-on experience building CI-style pipelines that compile, flash, and validate software on physical hardware targets. Skilled in test case development, automated regression testing, and hardware-in-the-loop (HIL) validation using C and Python.

Education

University of British Columbia **Expected May 2028**
*BASc in Electrical Engineering (**Co-op**, available September 2026)* *Vancouver, BC*

Experience

Norsat **June 2025 – August 2025**
Hardware Test Intern *Richmond, BC*

- Developed and executed structured test cases for system-level validation, ensuring software and hardware behavior met defined specifications.
- Automated data acquisition and sweep procedures to improve test repeatability and reduce manual validation time across the software build cycle.
- Diagnosed system-level failures through log analysis, contributing to improved production yield across 50+ units.

UBC Formula Racing Design Team **September 2025 – Present**
Electrical & Firmware Project Lead *Vancouver, BC*
GitHub

- Leading firmware development and CI workflow for an embedded sensing system, integrating build automation with real-time deployment and validation on physical hardware targets.
- Defined test cases and validation thresholds to verify firmware correctness under varying operating conditions, enabling repeatable regression testing across software iterations.
- Designed mixed-signal hardware with emphasis on reliable firmware-hardware interfacing, supporting continuous software integration onto the target platform.

Software & Embedded Projects

Autonomous Robot Controller | *Embedded C, STM32, CI/CD, HIL* **April 2026**
GitHub

- Built a Makefile-based CI-style pipeline to automatically compile, flash, and validate firmware behaviour on STM32 hardware targets for each software iteration.
- Developed modular embedded C firmware with structured and staged test cases to verify software behavior during each integration cycle.
- Performed hardware-in-the-loop (HIL) testing to validate sensor integration and system correctness through iterative automated regression cycles.

Oven Reflow Controller | *A51 Assembly, Python, HIL Testing* **January 2026**
GitHub

- Designed bare-metal firmware with ADC feedback, PWM control, and FSM-based thermal profiles; implemented automated Python HIL test framework for continuous regression validation at $\pm 3°$C accuracy.
- Developed structured test cases covering edge-case thermal profiles and IR-based user interface events, enabling repeatable software validation on the physical hardware target.
- Integrated real-time display control and event-driven firmware to support automated pass/fail test evaluation during each firmware build cycle.

Skills

CI/CD & Testing: Build Automation, Hardware-in-the-Loop (HIL), Test Case Development, Automated Regression Testing, Software Deployment on Hardware/MCU Targets
Programming: C, C++, Python, Makefiles, A51 & RISC-V Assembly
Embedded Systems: STM32, ESP32, ARM Cortex, 8051, EFM8
Protocols: SPI, UART, I^2C, CAN, USB, RS-485
Tools: Git, Altium, Oscilloscope, Signal Analyzer, Multimeter